Modular Steering Wheel Input and Dashboard Display System

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ECE Senior Capstone Project, ECD 619

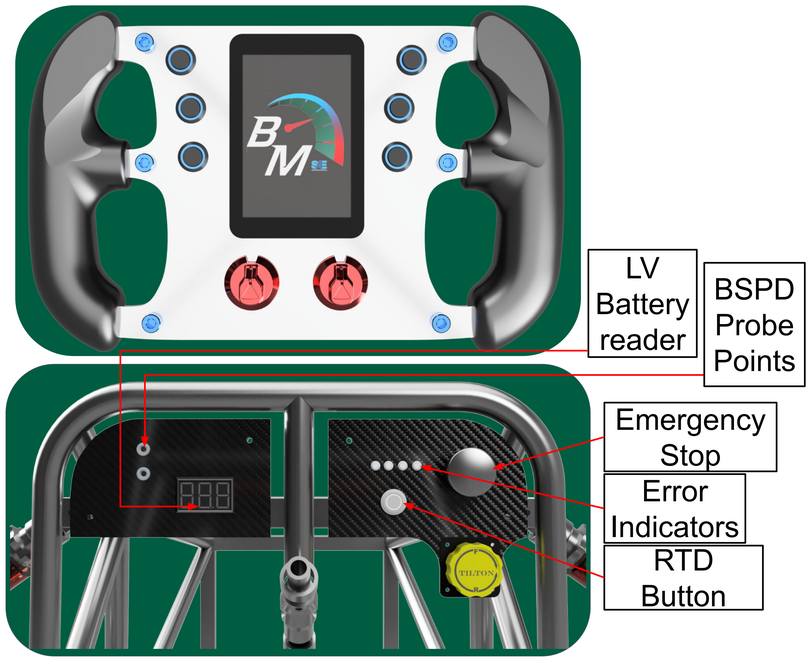
BINGHAMTON MOTORSPORTS

Introduction

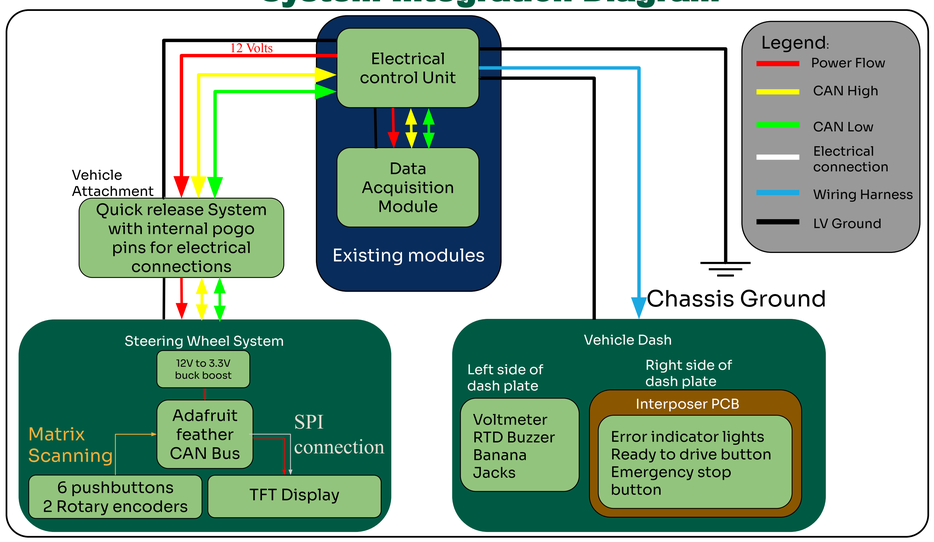
The motorsports team's previous steering wheel design established the foundation for driver-facing systems, with the dashboard running on a Raspberry Pi for development flexibility. The new design builds on this by integrating driver input controls and transitioning to a purpose-built microcontroller, improving boot times and real-time responsiveness. The result is a fully modular, low-latency system combining a live dashboard display, driver inputs, and onboard data acquisition into a single, competition-ready interface.

Key Requirements

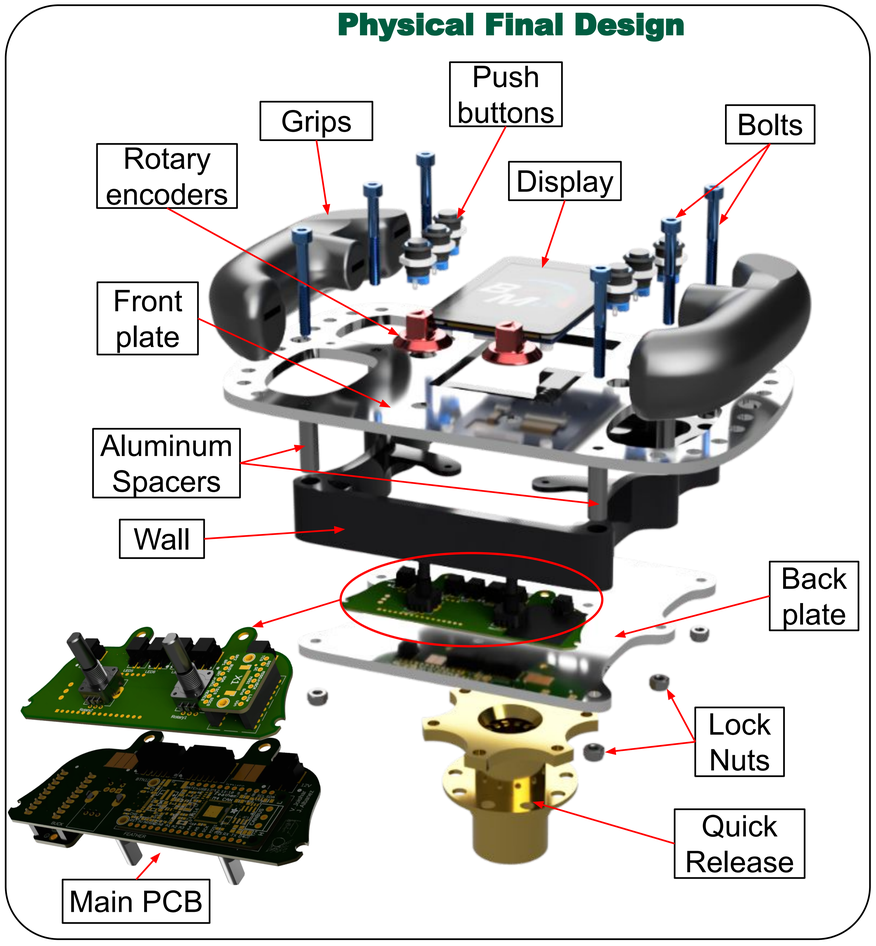
- CAN bus interface for BMS, inverter, and ECU signals
- TFT display for live telemetry, vehicle status, and diagnostics
- 2 rotary encoders and 6 push buttons on the steering wheel
- RTD and Emergency Stop buttons per FSAE EV rules
- Boot under 2 seconds, minimal operational latency
- Onboard data logging
- Dust- and water-resistant housings for both modules
- Reliable under vibration, thermal cycling, humidity, and EMI



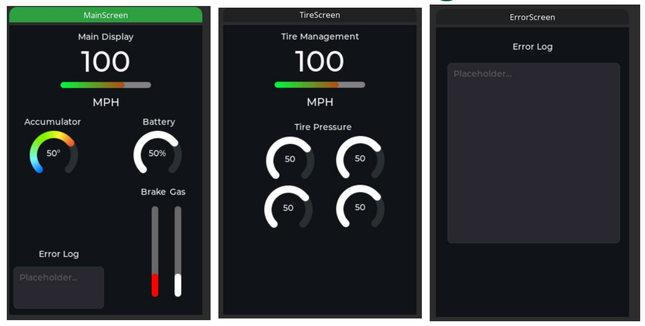
System Integration Diagram



Physical Final Design



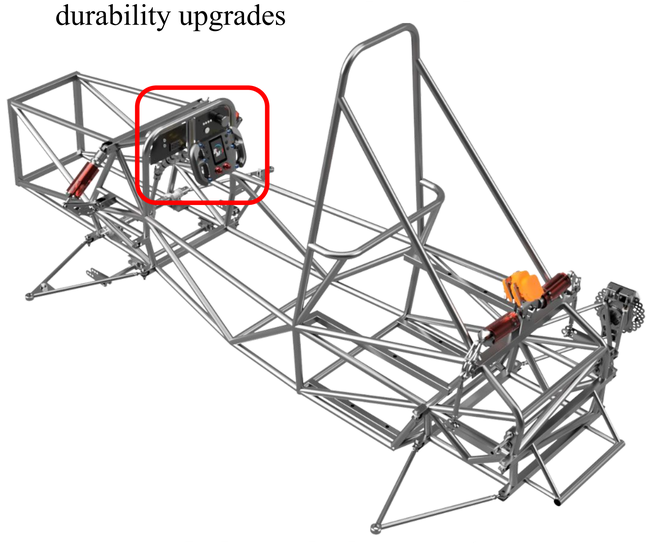
Software Final Design



- Coded with C and C++ using LVGL
- Receives and displays real-time ECU data via CAN BUS
- Logs data for testing and system verification
- Modular design to allow for new data to be displayed and new screens to be created
- Screens can be scrolled using the push buttons

Conclusion and Future Developments

- Low-latency steering wheel dashboard with fast boot and real-time response
- Integrated CAN, inputs, and display into a compact system
- Improved UI and advanced telemetry visualization planned
- Wireless communication, power optimization, and durability upgrades



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